

A Practical Guide to Algorithmic bias

TOPIC -- ALGORITHMIC BIAS

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Documentation and accountability frameworks Several documentation approaches have been proposed to improve transparency and support the evaluation of bias in algorithmic systems. One widely cited method is the use of model cards, which provide standardized summaries of an AI system's intended uses, performance metrics, evaluation datasets, and known limitations. Related efforts include datasheets for datasets, which outline the provenance, composition, collection methods, and recommended uses of training data. These documentation frameworks aim to clarify the assumptions and potential biases embedded in training data and machine-learning systems, helping practitioners, auditors, and impacted groups better interpret system behavior. In addition to documentation practices, researchers and policymakers have encouraged the use of structured governance mechanisms such as algorithmic impact assessments, risk-based evaluation procedures, and post-deployment monitoring. These processes seek to identify potential disparate impacts before deployment and ensure that AI systems continue to be evaluated for fairness during real-world operation. Public-sector initiatives such as Canada's Directive on Automated Decision-Making require impact assessments, explainability measures, and regular audits for certain high-risk automated systems. Together, these governance approaches complement technical mitigation strategies by embedding accountability and transparency throughout the lifecycle of AI development and deployment.

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Europe The General Data Protection Regulation (GDPR), the European Union's revised data protection regime that was implemented in 2018, addresses "Automated individual decision-making, including profiling" in Article 22. These rules prohibit "solely" automated decisions which have a "significant" or "legal" effect on an individual, unless they are explicitly authorised by consent, contract, or member state law. Where they are permitted, there must be safeguards in place, such as a right to a human-in-the-loop, and a non-binding right to an explanation of decisions reached. While these regulations are commonly considered to be new, nearly identical provisions have existed across Europe since 1995, in Article 15 of the Data Protection Directive. The original automated decision rules and safeguards found in French law since the late 1970s.

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Online hate speech In 2017 a Facebook algorithm designed to remove online hate speech was found to advantage white men over black children when assessing objectionable content, according to internal Facebook documents. The algorithm, which is a combination of computer programs and human content reviewers, was created to protect broad categories rather than specific subsets of categories. For example, posts denouncing "Muslims" would be blocked, while posts denouncing "Radical Muslims" would be allowed. An unanticipated outcome of the algorithm is to allow hate speech against black children, because they denounce the "children" subset of blacks, rather than "all blacks", whereas "all white men" would trigger a block, because whites and males are not considered subsets. Facebook was also found to allow ad purchasers to target "Jew haters" as a category of users, which the company said was an inadvertent outcome of algorithms used in assessing and categorizing data. The company's design also allowed ad buyers to block African-Americans from seeing housing ads. While algorithms are used to track and block hate speech, some were found to be 1.5 times more likely to flag information posted by Black users and 2.2 times likely to flag information as hate speech if written in African American English.

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Voting behavior A series of studies about undecided voters in the US and in India found that search engine results were able to shift voting outcomes by about 20%. The researchers concluded that candidates have "no means of competing" if an algorithm, with or without intent, boosted page listings for a rival candidate. Facebook users who saw messages related to voting were more likely to vote. A 2010 randomized trial of Facebook users showed a 20% increase (340,000 votes) among users who saw messages encouraging voting, as well as images of their friends who had voted. Legal scholar Jonathan Zittrain has warned that this could create a "digital gerrymandering" effect in elections, "the selective presentation of information by an intermediary to meet its agenda, rather than to serve its users", if intentionally manipulated.

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Speciesist bias Speciesist bias (also known as anthropocentric bias) refers to the tendency of large language models to systematically devalue or discriminate against non-human animals, often by prioritizing human interests or reinforcing the objectification of animals. This bias typically manifests as anthropocentrism, where the AI views animals primarily through their utility to humans (e.g., as food, tools, or pests) rather than as sentient beings with intrinsic value.

The earliest computer programs were designed to mimic human reasoning and deductions, and were deemed to be functioning when they successfully and consistently reproduced that human logic. In his 1976 book *Computer Power and Human Reason*, artificial intelligence pioneer Joseph Weizenbaum suggested that bias could arise both from the data used in a program, but also from the way a program is coded. Weizenbaum wrote that programs are a sequence of rules created by humans for a computer to follow. By following those rules consistently, such programs "embody law", that is, enforce a specific way to solve problems. The rules a computer follows are based on the assumptions of a computer programmer for how these problems might be solved. That means the code could incorporate the programmer's imagination of how the world works, including their biases and expectations. While a computer program can incorporate bias in this way, Weizenbaum also noted that any data fed to a machine additionally reflects "human decision making processes" as data is being selected. Finally, he noted that machines might also transfer good information with unintended consequences if users are unclear about how to interpret the results. Weizenbaum warned against trusting decisions made by computer programs that a user doesn't understand, comparing such faith to a tourist who can find his way to a hotel room exclusively by turning left or right on a coin toss. Crucially, the tourist has no basis of understanding how or why he arrived at his destination, and a successful arrival does not mean the process is accurate or reliable. An early example of algorithmic bias resulted in as many as 60 women and ethnic minorities denied entry to St. George's Hospital Medical School per year from 1982 to 1986, based on implementation of a new computer-guidance assessment system that denied entry to women and men with "foreign-sounding names" based on historical trends in admissions. While many schools at the time employed similar biases in their selection process, St. George was most notable for automating said bias through the use of an algorithm, thus gaining the attention of people on a much wider scale. In recent years, as algorithms increasingly rely on machine learning methods applied to real-world data, algorithmic bias has become more prevalent due to inherent biases within the data itself. For instance, facial recognition systems have been shown to misidentify individuals from marginalized groups at significantly higher rates than white individuals, highlighting how biases in training datasets manifest in deployed systems. A 2018 study by Joy Buolamwini and Timnit Gebru found that commercial facial recognition technologies exhibited error rates of up to 35% when identifying darker-skinned women, compared to less than 1% for lighter-skinned men. Algorithmic biases are not only technical failures but often reflect systemic inequities embedded in historical and societal data. Researchers and critics, such as Cathy O'Neil in her book *Weapons of Math Destruction*

(2016), emphasize that these biases can amplify existing social inequalities under the guise of objectivity. O'Neil argues that opaque, automated decision-making processes in areas such as credit scoring, predictive policing, and education can reinforce discriminatory practices while appearing neutral or scientific.